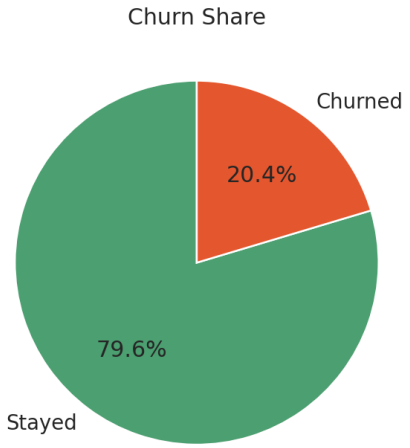
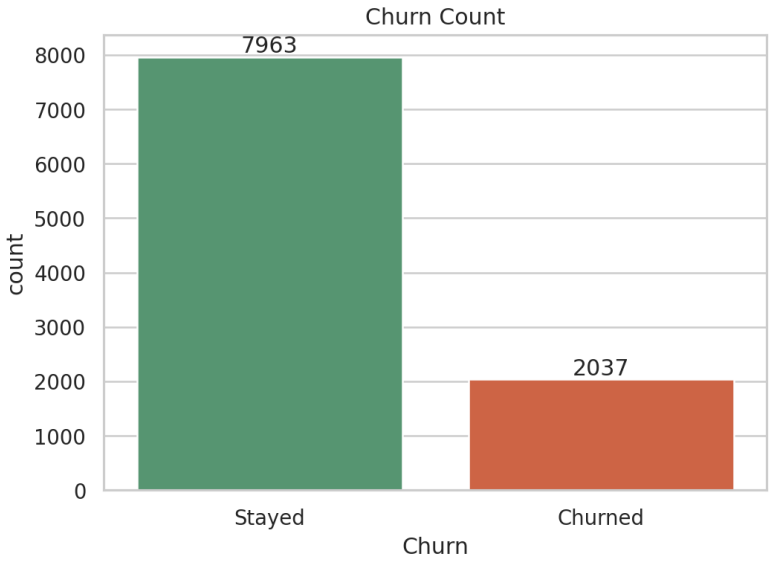
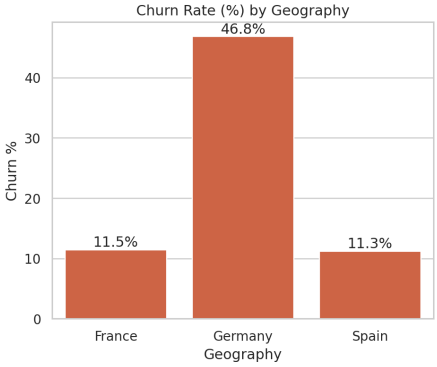
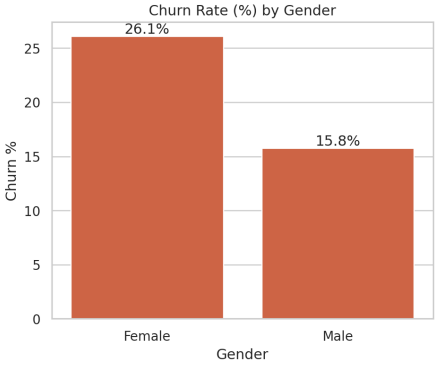
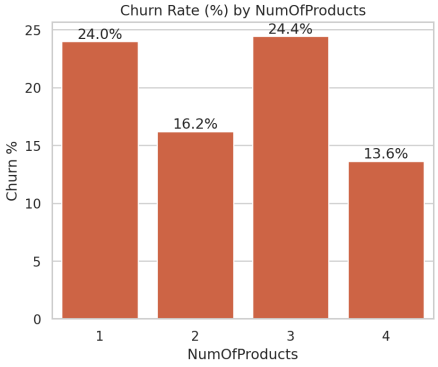
Bank Customer Churn Analysis — Visualization Graphs

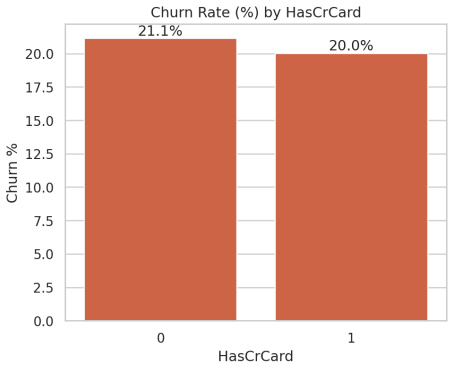


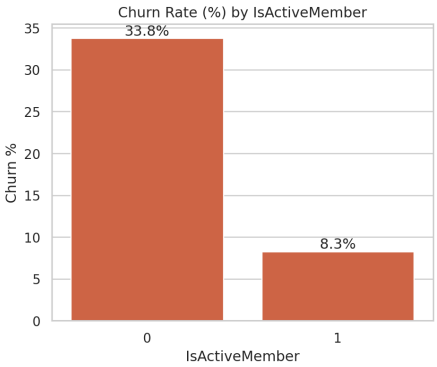


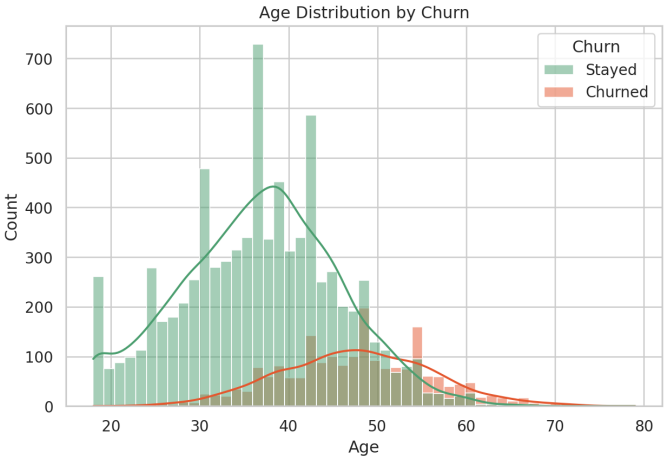

Churn Rate (%) by Geography
Churn %
Geography
France
Germany
Spain
11.5%
46.8%
11.3%

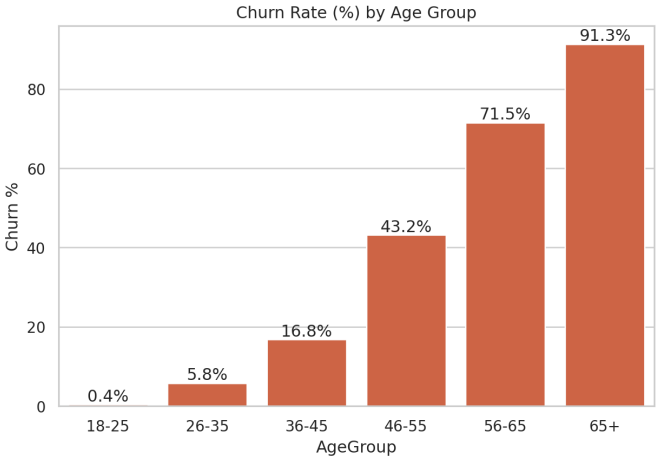

Churn Rate (%) by Gender
Churn %
Gender
Female
Male
26.1%
15.8%

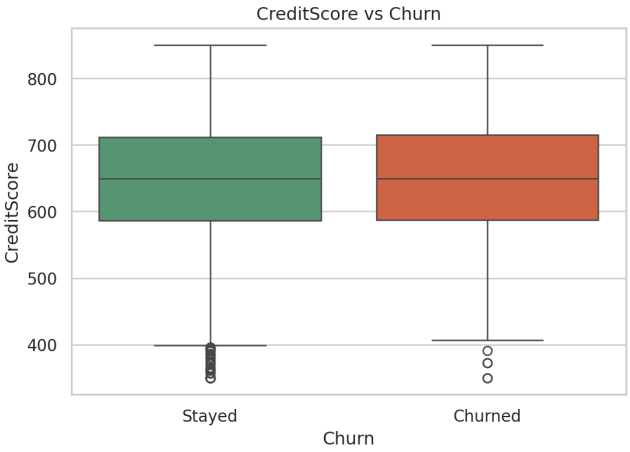

Churn Rate (%) by NumOfProducts
Churn %
NumOfProducts
24.0%
16.2%
24.4%
13.6%

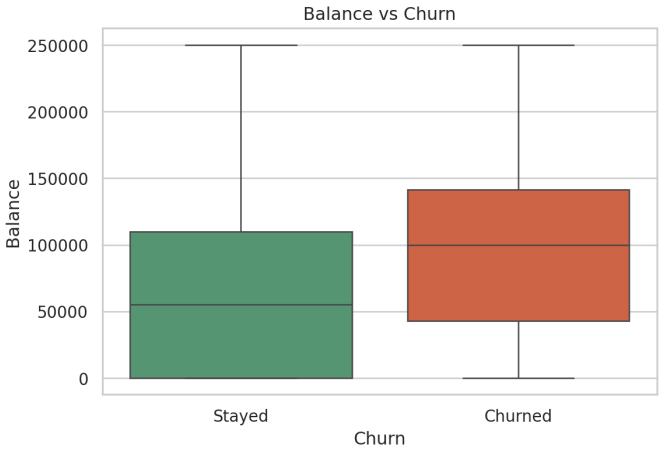

Churn Rate (%) by HasCrCard
Churn %
HasCrCard
21.1%
20.0%

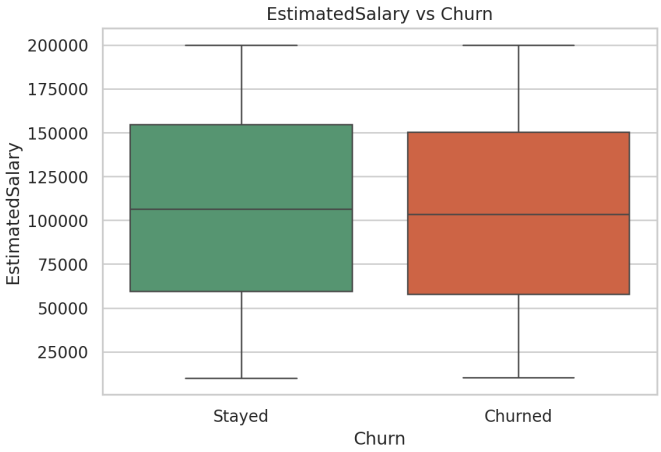

Churn Rate (%) by IsActiveMember
Churn %
IsActiveMember
33.8%
8.3%

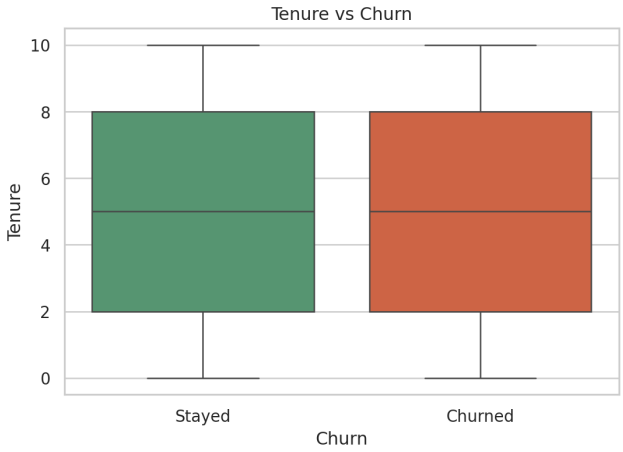

Age Distribution by Churn
Churn
Stayed
Churned
Count
Age
0
100
200
300
400
500
600
700
20
30
40
50
60
70
80

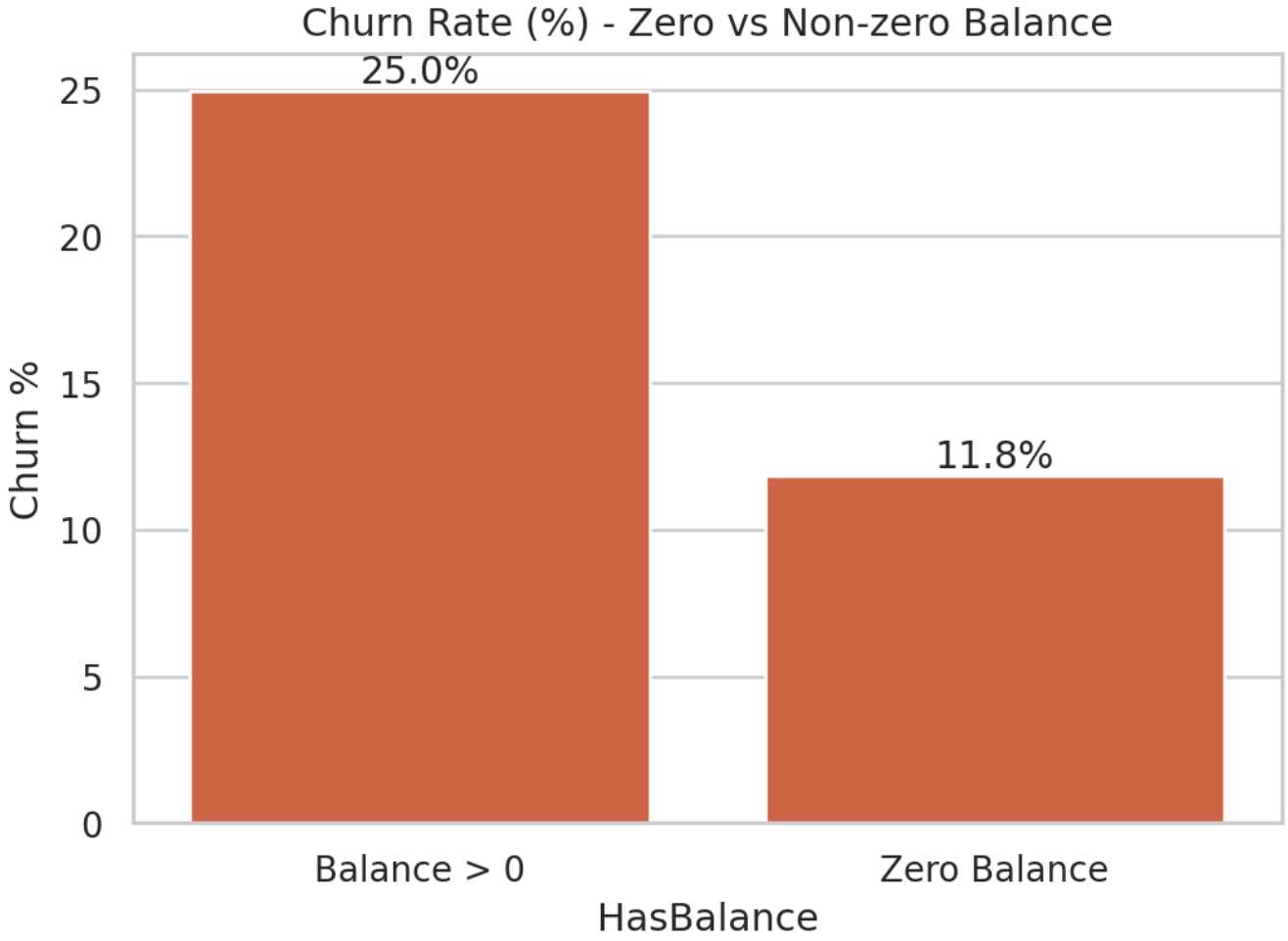

Churn Rate (%) by Age Group
Churn %
AgeGroup
0
20
40
60
80
0.4%
5.8%
16.8%
43.2%
71.5%
91.3%
18-25
26-35
36-45
46-55
56-65
65+

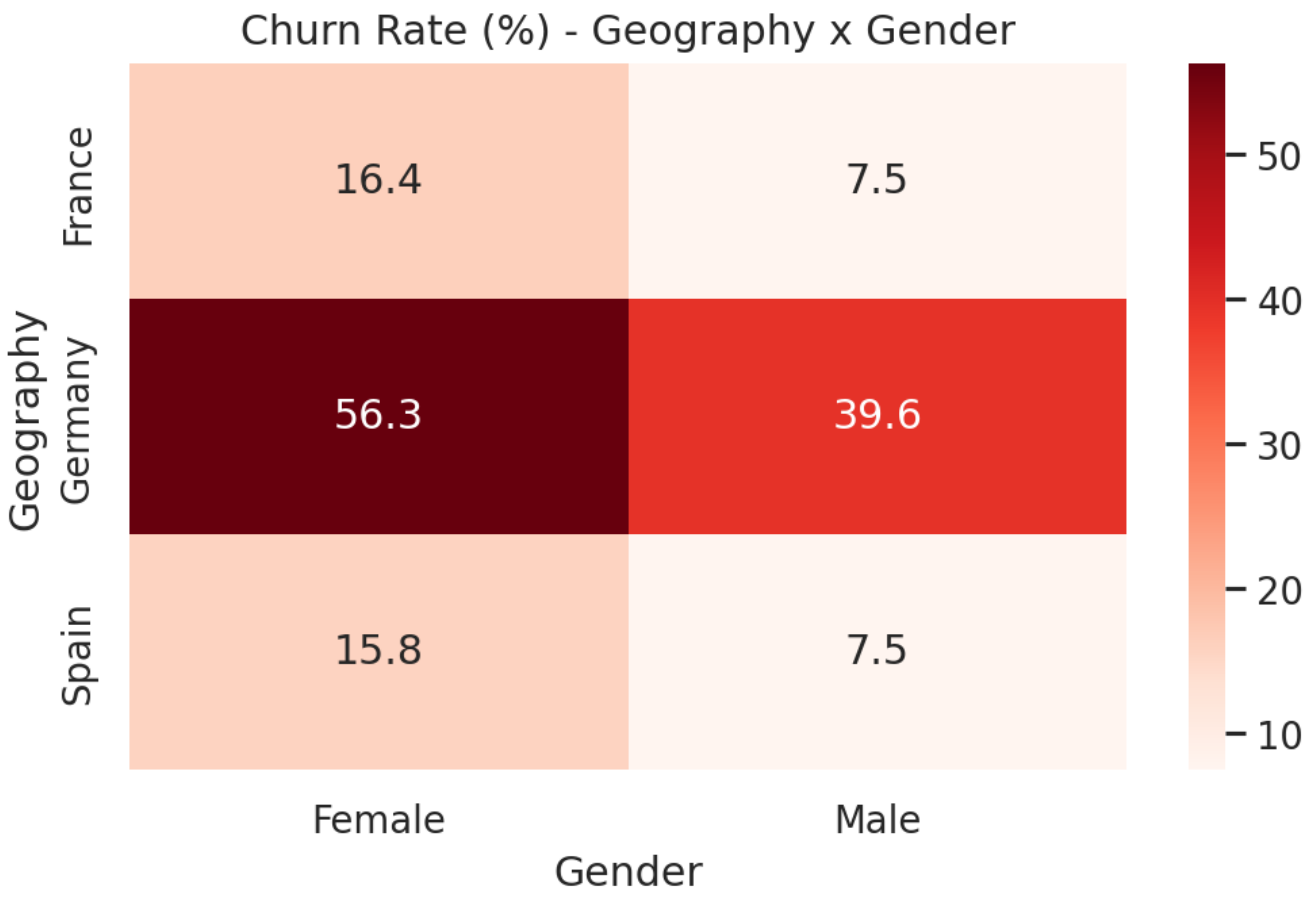

CreditScore vs Churn
CreditScore
800
700
600
500
400
Stayed
Churned
Churn

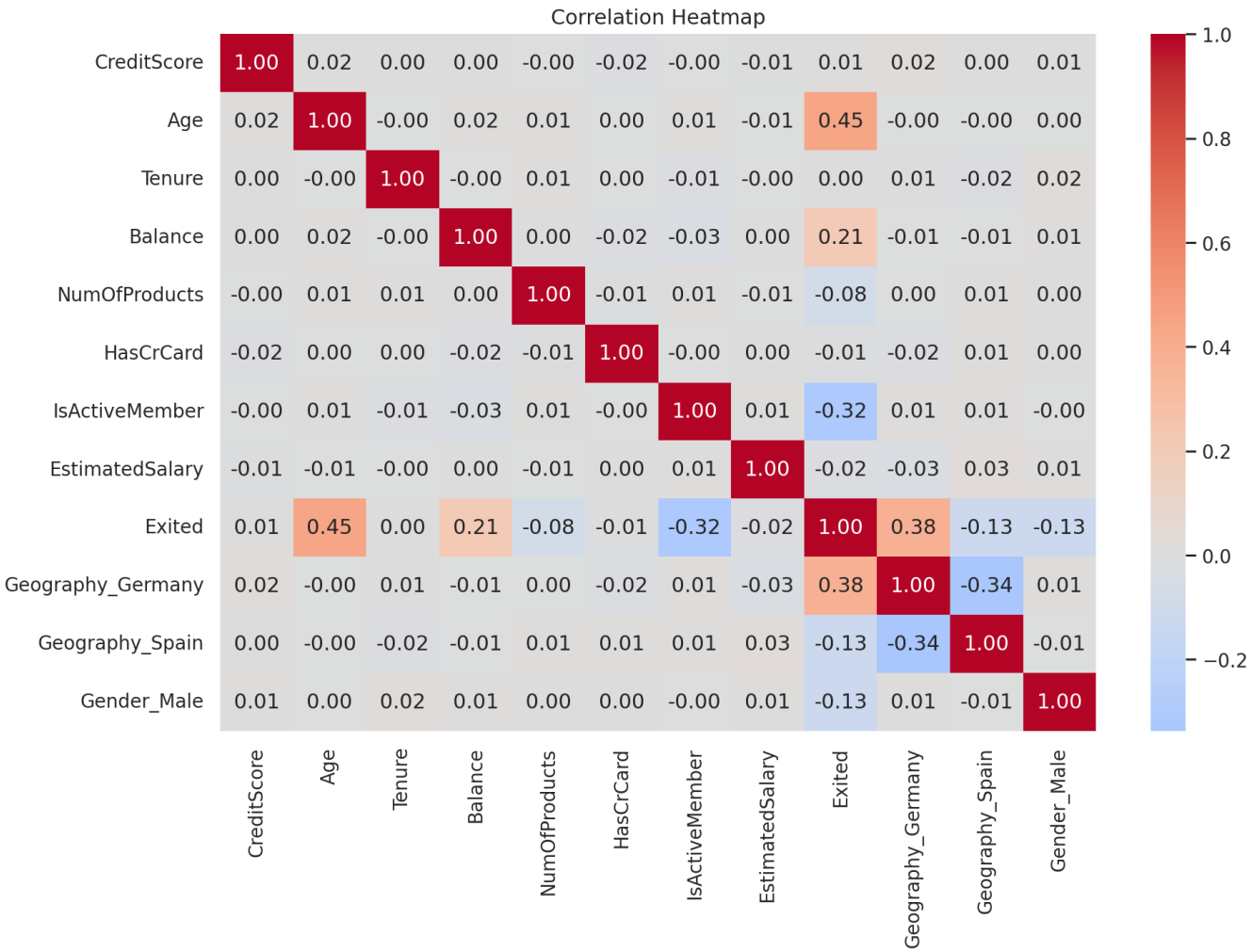

Balance vs Churn
Balance
250000
200000
150000
100000
50000
0
Stayed
Churned
Churn

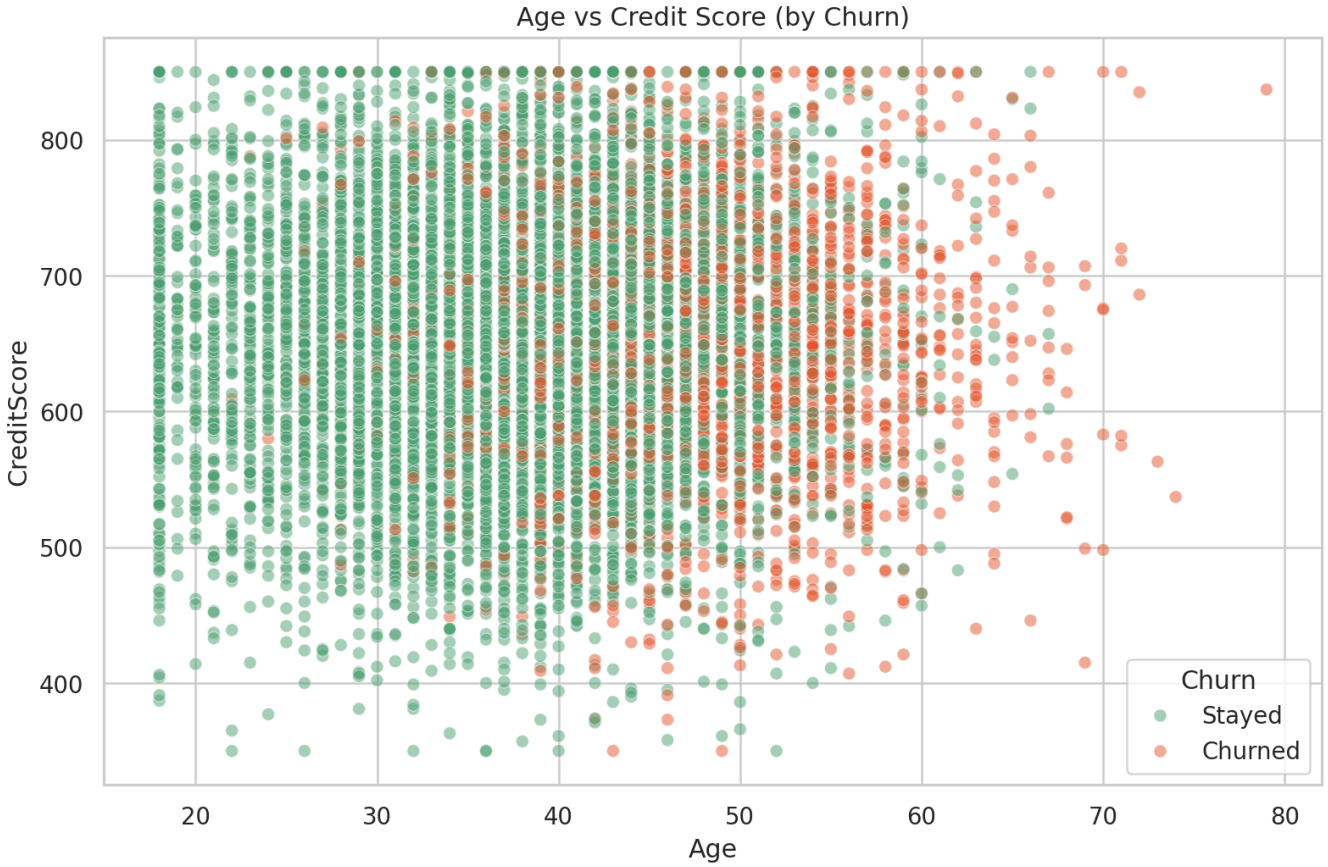

EstimatedSalary vs Churn
EstimatedSalary
200000
175000
150000
125000
100000
75000
50000
25000
Stayed
Churned
Churn


Tenure vs Churn
Tenure
10
8
6
4
2
0
Stayed
Churned
Churn


Churn Rate (%) - Zero vs Non-zero Balance
25.0%
11.8%
Churn %
0
5
10
15
20
25
Balance > 0
Zero Balance
HasBalance


Churn Rate (%) - Geography x Gender
Geography
France
Germany
Spain
16.4
7.5
56.3
39.6
15.8
7.5
Female
Male
Gender
50
40
30
20
10


Correlation Heatmap
CreditScore 1.00 0.02 0.00 0.00 -0.00 -0.02 -0.00 -0.01 0.01 0.02 0.00 0.01
Age 0.02 1.00 -0.00 0.02 0.01 0.00 0.01 -0.01 0.45 -0.00 -0.00 0.00
Tenure 0.00 -0.00 1.00 -0.00 0.01 0.00 -0.01 -0.00 0.00 0.01 -0.02 0.02
Balance 0.00 0.02 -0.00 1.00 0.00 -0.02 -0.03 0.00 0.21 -0.01 -0.01 0.01
NumOfProducts -0.00 0.01 0.01 0.00 1.00 -0.01 0.01 -0.01 -0.08 0.00 0.01 0.00
HasCrCard -0.02 0.00 0.00 -0.02 -0.01 1.00 -0.00 0.00 -0.01 -0.02 0.01 0.00
IsActiveMember -0.00 0.01 -0.01 -0.03 0.01 -0.00 1.00 0.01 -0.32 0.01 0.01 -0.00
EstimatedSalary -0.01 -0.01 -0.00 0.00 -0.01 0.00 0.01 1.00 -0.02 -0.03 0.03 0.01
Exited 0.01 0.45 0.00 0.21 -0.08 -0.01 -0.32 -0.02 1.00 0.38 -0.13 -0.13
Geography_Germany 0.02 -0.00 0.01 -0.01 0.00 -0.02 0.01 -0.03 0.38 1.00 -0.34 0.01
Geography_Spain 0.00 -0.00 -0.02 -0.01 0.01 0.01 0.01 0.03 -0.13 -0.34 1.00 -0.01
Gender_Male 0.01 0.00 0.02 0.01 0.00 0.00 -0.00 0.01 -0.13 0.01 -0.01 1.00
CreditScore
Age
Tenure
Balance
NumOfProducts
HasCrCard
IsActiveMember
EstimatedSalary
Exited
Geography_Germany
Geography_Spain
Gender_Male
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


Age vs Credit Score (by Churn)
CreditScore
800
700
600
500
400
20
30
40
50
60
70
80
Age
Churn
Stayed
Churned